

Engineering Report - Q3 2024

Prepared by the Engineering Planning Office for the steering committee. This document summarizes the deliverables of the third quarter, the resource usage, and the outlook for Q4.

1. Delivered work packages

Nine of eleven planned work packages shipped in Q3. The two deferred packages (WP-04 telemetry gateway, WP-09 field calibration kit) were moved to Q4 with an approved scope change.

WP	Title	Effort (h)	Status
WP-01	Firmware 2.4 baseline	320	Done
WP-02	Power supply redesign	410	Done
WP-03	Sensor calibration rig	260	Done
WP-05	HMI software 3.1	380	Done
WP-06	EMC pre-compliance campaign	190	Done
WP-07	Packaging redesign	120	Done
WP-08	Field trial documentation	140	Done
WP-10	Spare parts portfolio	90	Done
WP-11	Service documentation	230	Done

2. Resource usage

Engineering capacity was 2,140 hours in Q3; 1,930 hours were booked against work packages, 120 hours went to support of the production ramp, and 90 hours were unassigned. Overtime stayed within the approved envelope of 5 %.

3. Risks and mitigations

The supply chain for the new power module remains the top risk. The second-source qualification is under way; the target date for full second-sourcing is 2025-02-01. The EMC campaign confirmed margins above 6 dB on all conducted and radiated tests.

The HMI software team reported a usability finding from the field trial regarding the alarm list; the fix is scheduled for the 3.1.2 maintenance release in October.

4. Outlook for Q4

Q4 focuses on the two deferred packages, the production ramp of the redesigned power supply, and the preparation of the 2025 platform review. A mid-quarter checkpoint is scheduled for calendar week 46, and the final steering committee meeting of the year takes place on 2024-12-17.

The engineering planning office will publish the Q4 work package breakdown by 2024-10-08. Any remaining questions can be directed to the program office; the escalation path is unchanged from the Q2 report.

Approved by the engineering director on 2024-10-02, revision A. The next report in this series covers Q4 2024 and is due on 2025-01-10.

5. Detailed notes for the steering committee

Field trial feedback on the HMI alarm list has been triaged into three severity classes. Class A items block ship acceptance and were closed in the 3.1.2 maintenance release. Class B items ship with workarounds and will be closed in 3.1.3. Class C items remain backlog for the 2025 platform review and do not affect the current production ramp.

The production ramp of the redesigned power supply is sequenced in three waves. Wave 1 covers the Stuttgart pilot line, wave 2 the main assembly hall, and wave 3 the spare-parts and training fleet. Each wave has an independent acceptance checklist that the quality office signs before the next wave starts.

Documentation packages for service and field trial use a common template so that revision tracking stays consistent across languages. The translation backlog for French and Spanish is tracked separately and is not on the critical path for

the December steering meeting.
Capacity planning for early 2025 assumes a steady 2,100 engineering hours per quarter and a 4 % overtime ceiling.
Hiring for two firmware engineers is open; until those seats are filled, the deferred telemetry gateway package remains the primary schedule risk after second-source qualification of the power module.

Stakeholders who need the raw effort tables or the full risk register can request them from the program office. This report intentionally keeps the narrative form used in prior quarters so that year-on-year comparison remains straightforward for the committee.